ACP CW3 / CW4 Specifications

28.02.2026

Background information (details to follow later in the document):

- **CW3** provides an opportunity for you to create anything you consider important as an addition to and in the context of ACP.

  You are free to use AI, any language of choice and any environment.

  The marking will be based on a submitted document cw3_explanation.pdf in PDF format not exceeding 1,000 words and 30 points are available

  Important: Without this PDF explanation and the submitted sources to build your solution CW3 cannot be marked and will be considered 0 points.

- **CW4** is the video presentation for CW3 where you showcase your solution with a short demo, some background and some implementation details.

  Video length is 8-11 minutes. 15 points are available. The rubric can be found later in this document

  Important: Without a CW3 submission CW4 cannot be marked and will be considered 0 points.

- **To keep everything in a realistic setting do not attempt excessive workloads for yourself. There is always another feature and one more thing to do – limit and plan so that you don't deplete your reserves too much.**

- **The Rubrics for CW3 / 4 will be delivered in a separate document**

AI usage is allowed in full for CW3, and as a conceptual tool in CW4 (no content generation!).

CW3

In CW3 you can implement whatever you want – if it is in the range of ACP. Use AI, any language you want and implement anything you consider useful.

More complex solutions could potentially gain higher points, yet this is not going to be certain as sometimes very innovative small tools and additions provide huge benefits.

You can extend the CW2 implementation or create something entirely new. Just submit everything you did and – please... - some instructions on how to build it.

The anticipated time for this task will be 12 - 16h, so, quite limited. Please, try not to get overboard by investing too much time – always consider the points and how much improvement those extra 10h will make.... We anticipate that solutions will be more a "proof of concept" (PoC) than complete implementations with all bells and whistles. Thus, do not try to create the richest, most complete implementations, but look for innovative ideas and use cases, things which are complex and can easily be adjusted, etc.

It could be (just examples – use your imagination):

- Using Flink to analyze some real-time data
- Using DynamoDB integrating with queues
- S3 + Lambdas + Queues
- Postgres with Kafka
- ...

Together with the submission of your work you must submit a PDF document cw3_explanation.pdf which explains (based on the rubrics later in this document) why you did what for whom and which problem to you solve. In addition, you explain how you did it and what was your motivation for the tools used...

Please upload all necessary to re-create your submission – if you rely on tools / services a reference is fine (so, no upload of external images, etc.)

The document must not exceed 1,000 words maximum. For each 100 words excess, 10% will be deducted and if more than 1,500 words, 0 will be awarded as a mark. Use graphics, charts, etc. to make it better readable; footnotes and endnotes leave a good impression as well... After all, it's an essay.

Your marking (30 points) will be based on the following rubrics :

- Innovation / Idea / Benefit (12)

What makes your idea unique, new, who benefits, why is it important

- Execution / Implementation (8)

  How well did you use the tools and technologies. Was it "just" a bit of thrown together pieces or a more complex, properly engineered implementation.

- Completeness (6)

  How much did you cover of your idea. Did you provide a "full" feature or a PoC

- Style (4)

  How is the style of the implementation. Hobbyist, Professional? Does it feel "mature"

Your PDF should thus address these items (more can be done in the video) accordingly. In addition, you should mention how you what AI you used, how it influenced the work, etc.

In principle, it is important to understand why you did what, for whom, to solve which problem with which attempt / solution / technology and how.

CW4

In CW4 you will create a video showcasing the solution you built in CW3. The marking (15 points) will be based on the rubrics and in principle you express in your own words what you built, why, for whom, how and with which tools.

The following criteria will be applied:

- Background elaboration / target audience (2)

  Explain for whom this is important and which problem domain you are working in

- Problem statement (3)

  What did your solution tackle? Why does it help / improve?

- Explanation of the student's choice (5)

  How well can you explain why you did what and how you did it?

- Presentation (5)

  Use of animation, gestures, speech, presentation, etc. How good is the video to watch? 11 min of text only, or interactive, good "pick the audience up"?

The main part shall be the presentation of what you did as a demo.

Your video is 9-11 minutes long with 10% off for each minute over and 0 as mark for videos exceeding 11 minutes. If you manage all you want to say in 6, this is as well – usually it is easier for students to talk longer, so no rush, if you are faster.

Some important instructions for the video:

- **MP4-format (MOV or AVI only in exceptional cases)**
- In the very beginning your face must be visible and next to it your student id. Say your student id for us
- Then use your left hand, spread your fingers and move before your face left to right and back
- Same with right hand
- It is important that your face is visible through your fingers